

DIFFUSION IN SULFIDES

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INTRODUCTION

The sulfides are widely distributed in nature and are used in many different ways in commercial applications. There are correspondingly diverse reasons for trying to understand their properties. Ferrous sulfide is a good example of a substance for which effort from several disciplines yielded a wealth of related information. Geologists investigated the phase boundaries intensively when there was hope that the pyrite-pyrrhotite equilibrium, preserved by natural quenching, might serve as a geological barometer and thermometer. Solid-state physicists, especially in Japan, studied the unusual electronic and magnetic properties of ferrous sulfide. Crystallographers found the low-temperature transitions in electron-spin orientations and point-defect structures worth considerable effort. Metallurgists have been concerned for decades with the unfortunate effects of ferrous sulfide inclusions on the ductility of steels. The fast growth of the sulfide on iron in sulfur-containing environments at elevated temperatures, a problem of some magnitude in the petroleum industry, kindled my interest in this substance.

In place of a detailed review of the work of my students on ferrous sulfide, I shall survey more generally diffusion in sulfides, treating ferrous sulfide as one of several examples. The reader is also re-

ferred to the recent review of diffusion in the chalcogenides of Zn, Cd, and Pb by Stevenson (1973).

DIFFUSION IN SULFIDES COMPARED WITH DIFFUSION IN OXIDES

Although diffusion in oxides is still poorly understood, much more is known about oxides than sulfides. A general comparison has some value mainly because several aspects of the influence of defect structure on diffusion seem to be defined more clearly in the sulfides than in the oxides.

One difference affecting ion mobilities in these compounds is that the sulfide ion is much larger than the oxide ion. Moreover, the larger sulfide ion has a polarizability more than three times as large as the oxide ion. Therefore, the smaller cations occupy larger and softer interstices in the sulfides. Identifying the interstices as octahedral and tetrahedral sites if bounded by six or four anions, respectively (regardless of small distortions from regular packing), the constrictions in the channels connecting these sites also are wider and softer in the sulfides. The strains necessary to squeeze cations through these barriers into empty sites determine the activation energies for diffusion. Hence, lower activation energies can be expected for sulfides than for the corresponding oxides. Table 1 lists

TABLE 1. Ionic radii in angstrom units for selected ions

Cu ⁺ 0.96	Fe ²⁺ 0.74	Fe ³⁺ 0.64	Q ⁵⁺ 1.40	F ⁻ 1.36
Zn ⁺ 0.88	Cu ²⁺ 0.72		S TM 1.84	Cl ⁻ 1.81
Ag ⁺ 1.26	Zn ²⁺ 0.74		Se ⁵⁻ 1.98	Br ⁻ 1.95
Cd ⁺ 1.14	Ag ²⁺ 0.89		Te [*] ~2.21	I TM 2.16
Au ⁺ 1.37	Cd ¹⁺ 0.97	An ⁸⁴⁺ 0.85		

ionic radii for a few anions and cations considered in this account plus a few extras for comparison.

Diffusion in crystals depends on the motion of point defects. The sulfides show two related types of disordering more clearly than the oxides. Bertaut (1956) determined the structure of the vacancy superlattice in the monoclinic phase that has a composition near Fe_7S_8 . At higher temperatures the vacant cation sites are more randomly distributed, but a strong short-range interaction or clustering must persist to account for the lower mobility of these vacancies compared with the mobility of the vacancies near the FeS composition. A similar dependence of vacancy mobility is found in highly nonstoichiometric ferrous oxide where x-ray measurements have demonstrated the presence of vacancy clusters (Roth, 1960, and Koch and Cohen, 1969).

SILVER AND COPPER SULFIDES

The crystal structure usually defines either the octahedral (e.g., NaCl structure) or tetrahedral sites (e.g., zincblende structure) as the structure sites. The other sites then become interstitial sites through which the ions may jump, or sometimes must jump, in the course of a unit diffusion process. If the ions enter the interstitial sites and jump from one such site to another without returning to structure sites often, the mechanism is called interstitial. Cation diffusion by interstitial mechanisms probably is more frequent in sulfides than in oxides because of the size effect. If the interstitial ion displaces an ion from a structure site into a neighboring interstitial site, the mechanism is interstitialcy. If the motion largely consists of an ion jumping from a structure site into a neighboring vacant structure site, perhaps passing quickly through an interstitial site, it is a vacancy mechanism. However, $\alpha\text{Ag}_2\text{S}$, the cubic form, has silver ions distributed almost at random over the

octahedral and interstitial interstices (Rahlf, 1953). In such a case there is no distinction between interstitial and vacancy mechanisms. The jump from one site to another requires a very small activation energy. In a sense, the cation part of the structure has melted while the anion part remains nearly intact.

Figure 1 shows the cation diffusivities for silver tracers in $\alpha\text{Ag}_2\text{S}$ (Allen and Moore, 1959 and Okazaki, 1967). The activation energy reported by Allen and Moore is only 3.45 kcal per gram mole, somewhat lower than the activation energy required for the silver ion to diffuse in aqueous solution at room temperature. Below 177°C , the monoclinic $\beta\text{Ag}_2\text{S}$ has lower diffusivities and an activation energy of about 10.5 kcal per gram mole. The mechanism appears to be predominantly vacancies on silver ion sites.

This sequence of structures, disordered cations at high temperature, cations on characteristic sites with vacancies nearly randomly distributed at intermediate temperatures, and ordered cation sites at low temperatures; as in Fe_7S_8 seems to be a consistent pattern. Although the sequence probably represents a universal tendency, it is obscured by phase transformations, including melting, that intrude at different levels of cation disorder in the different sulfides and oxides.

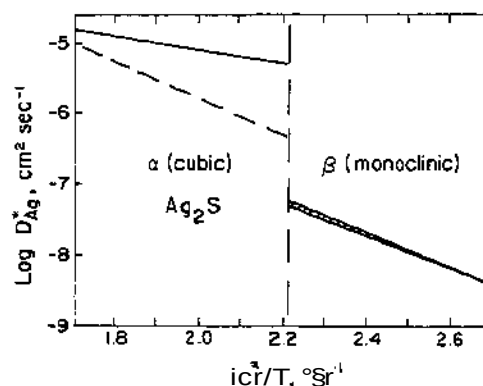


Fig. 1. Self-diffusion of silver in cubic (α) and monoclinic (β) Ag_2S . Solid lines, Allen and Moore (1959). Dashed line, Okazaki (1967).

At low temperatures the phase equilibria in the cuprous sulfides are complex (Mathieu and Rickert, 1972). Diffusion data are sparse and obtained mainly from sulfide growth measurements. It is interesting that the results of Etienne (1970) for low chalcocite and low digenite between 30° and 73°C differ from each other by only about a factor of two and lie about one order of magnitude below the extrapolation of the β -Ag₂S line. The results of Bartkowicz *et al.* (1969) for 350° to 550°C lie less than an order of magnitude above the extrapolation of the α -Ag₂S line. This striking similarity in diffusion behavior must reflect similar defect structures. The copper diffusion results are summarized in Table 2.

Sulfide ion diffusivities in silver sulfide have been shown to be much slower and to require much higher activation energies than cation diffusion. Ishiguro *et al.* (1953) reported¹

$$D_s = 2.4 \times 10^{-4} \exp \left(-24.0/RT \right) \text{ cm}^2\text{sec}$$

for the a (cubic form, while Peschanski (1950) reported

$$D_s = 0.24 \exp \left(-26.3/RT \right) \text{ cm}^2\text{sec}$$

for the p (monoclinic) form of Ag₂S.

¹ All activation energies in the exponential factors are in kcal per gm mole.

CADMIUM SULFIDE

CdS has useful photoconductive properties. In combination with Cu₂S it also may form useful photovoltaic devices. Pure CdS has n-type electronic behavior because of interstitial cadmium ions. The cadmium self-diffusivity has been measured several times, most recently by Jones (1972). He observed qualitatively the same features that others found: (1) Self-diffusivity of cadmium increases with cadmium pressure or decreases with increasing sulfur pressure, consistent with a cadmium interstitial defect model. (2) Cadmium self-diffusivity can be increased by doping with acceptor impurities. (3) Although the sulfide is hexagonal (wurtzite structure) there is little or no variation in diffusivity parallel versus perpendicular to the c axis. (4) A small proportion of fast-diffusing tracer ions runs well ahead of the remainder. The slower tracer ions are presumed to exchange with ions from structure sites to become temporarily immobilized. Excluding the small, fast-diffusing component, Jones found between 602° and 1255°C in excess cadmium vapor

$$D_{\text{Cd}}^* = 1.2 \exp \left(-53 \pm 5/RT \right) \text{ cm}^2\text{sec}$$

Figure 2 summarizes the published self-diffusion data for cadmium and sulfur in cadmium sulfide that is not doped

TABLE 2. Diffusion coefficients for copper in cuprous sulfides

T°C	D cm ² /sec		Reference
	<i>low temperature chalcocite</i>	<i>low temperature digenite</i>	
30	3.1×10^{-11}	1.5×10^{-11}	Etienne (1970)
73	2.8×10^{-10}	1.2×10^{-10}	
	$D_0 = 8.1 \times 10^{-10}$	3.6×10^{-10}	
	$Q = 5.87^*$	6.10^*	
	<i>high-temperature chalcocite</i>		
350	8.7×10^{-10}		Bartkowicz <i>et al.</i> (1969)
450	2.7×10^{-9}		
550	4.0×10^{-9}		
	$D_0 = 3.3 \times 10^{-9}$		
	$Q = 7.30^*$		

*kcal/g mole.

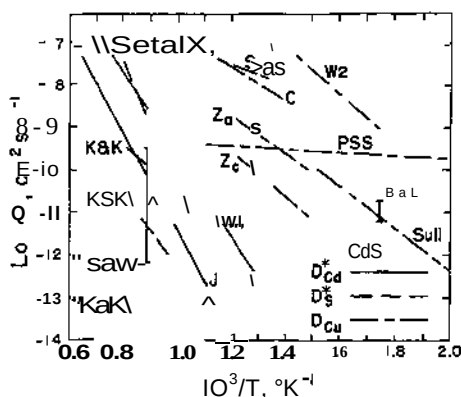


Fig. 2. Self-diffusion of cadmium and sulfur and diffusion of copper in CdS.

(W) Woodbury (1964). Under saturated cadmium vapor. At 800°C using sulfur vapor as well, self-diffusion of cadmium was linearly dependent on the donor impurity concentration.

(S&W) Shaw and Whelan (1969). The upper limit of the range (vertical line) at 850°C is for saturated cadmium vapor, the lower limit is extrapolated to saturated sulfur vapor. The self-diffusion coefficient, which was measured along the hexagonal axis had a plateau from P_{en} of 0.1 to 10^2 torr.

(K&K) Kumar and Kroger (1971). Self-diffusion of cadmium was measured under 1 atmosphere pressure of cadmium vapor. Self-diffusion of sulfur was measured under 2 atmospheres of sulfur pressure (upper dashed line) and 4 atmospheres of cadmium pressure (lower dashed line).

(J) Jones (1972). Self-diffusion of cadmium, under saturated cadmium vapor (to 1280°K) and excess cadmium beyond, showed no dependence on crystallographic orientation. In excess sulfur vapor the cadmium diffusivities were about 10-fold lower.

(S *et al*) Sysoev *et al* (1969). Sulfur self-diffusion in CdS. Crystals grown from the melt under inert gas pressure.

(€) Clarke (1959).

(W2) Woodbury (1965).

(S&S) Szeto and Somorjai (1966). Observed a fast and a slow component and postulated interstitial-substitutional equilibrium distribution.

(PSS) Purohit *et al*, (1969). Used CdS crystals clipped into cuprous chloride solution to form a Cu₂S layer on the surface.

(Sull) Sullivan (1969).

(Z., Zr) Zmija (1971), and Zmija and Demianiuk (1971). The subscripts refer to diffusion in the *a* and *c* directions, respectively.

(BdL) Birchenall and Lu, unpublished data.

intentionally. Some of the variations in the results reflect the sensitivity to the

sulfur or cadmium activity. (A similar dependence on sulfur activity was not observed in the more highly disordered Ag₂S.) However some of the variations, especially in the slopes, reflecting activation energies are undoubtedly experimental disagreement which may be due in some measure to impurities in the single crystals that were used. The references and some additional information about the measurements are given in the extended caption.

CdS with a layer of Cu₂S forms a *p-n* photovoltaic junction that is potentially useful as a solar energy converter. Diffusion of Cu into CdS appears to be essential in forming the *p-n* junction within the CdS. Further diffusion of Cu in CdS may lead to a decrease in photocell properties. Numerous measurements of bulk diffusivity have been made. They are included in Fig. 2 for direct comparison with Cd and S self-diffusion. The most precise measurements to the lowest temperature seem to be those of Sullivan (1969) by an indirect capacitance method that detected the width of the insulating intrinsic layer bounded by the *p-n* junction and the CdS-Cu₂S interface. His method does not detect the fast-interstitial tail observed by others (Zmija, 1971 and Zmija and Demianiuk, 1971).

We have been concerned that copper might diffuse rapidly along CdS grain boundaries to form an insulating layer, isolating whole grains from the solar cell. Copper was vapor deposited on coarse-grained polycrystals² of CdS and diffused for 24 hours at 300°C. A slightly beveled surface exposed the near-surface region to depths of 0 to about 30 microns. Microprobe scans across isolated grain boundaries revealed no excess copper in those regions. The most useful results

² I am grateful to Mr. Ludewig Vanderberg for growing these crystals and to Mr. Tien-Lien Lu for carrying out the diffusion measurements. Microprobe measurements of the copper in CdS were made by Mr. James Ficoa at Micron, Inc. Details of this work will be reported WSPH work in progress *in* completed.

came from the regions around pores, filled with vapor-deposited copper, from which grain boundaries emerged. Microprobe traces normal to the surfaces of such pores permitted calculation of a bulk diffusion coefficient for copper in cadmium sulfide, ignoring the fast-diffusing tail, that falls on Sullivan's line at 300°C. What our method lacks in precision it makes up in directness. Microprobe traces parallel to pore surfaces intersecting grain boundaries between 5 and 10 microns from the pore again showed no excess copper at the boundaries.

At 400°C Zmija found that for impurity diffusion in CdS, $D_{Cn} \gg D_{\pm g} > D_{Au}$, which is in the order expected, the smallest ion diffusing most easily, presumably by an interstitial mechanism.

FERROUS SULFIDE

Ferrous sulfide (Fe_{1-x}S) has the hexagonal nickel arsenide structure above about 300°C. Both iron and sulfur diffuse about twice as fast parallel to the *c* axis as perpendicular to it. Nickel sulfide shows a smaller enhancement in the *c* direction, and its diffusion coefficients have nearly the same magnitudes as those in ferrous sulfide (Klotsman, *et al.*, 1966).

It is possible to use the magnitude of the anisotropy and its virtual independence of composition and temperature to investigate the relative ease of atom jumps along the various channels in the structure (Condit, 1969). In the case of ferrous sulfide it appears that jumps seldom start and end in neighboring sites on the same basal plane and take place between planes by jumping from a normal (octahedral) site into an interstitial (tetrahedral) site instead of jumping directly along the *c* axis between two octahedral sites. The notion that diffusion both parallel and perpendicular to the *c* axis takes place at rates limited by the same sort of ion jump, from an octahedral site into a tetrahedral site which has a vacancy in the adjoining basal

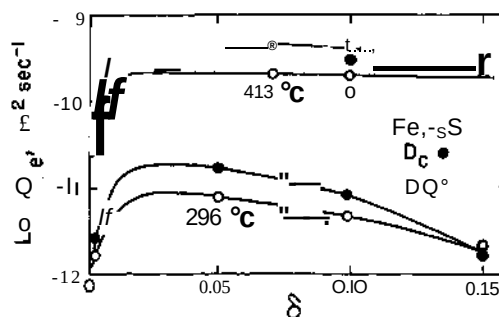


Fig. 3. Self-diffusion of iron in ferrous sulfide single crystals parallel (filled points) and perpendicular (open points) to the hexagonal (*c*) axis. Unpublished data by R. H. Condit, R. R. Hobbins, Jr., and C. E. Birchenall (submitted to *Oxidation of Metals*).

plane, is consistent with the observation that the activation energy is the same for both orientations.

Figure 3 shows that at 413°C, as at all temperatures investigated between 352° and 697°C, the iron ion diffusivity rises rapidly with the first increase in δ , to about 0.02, then practically levels off as δ goes to 0.15. This leveling of *D* corresponds to a decrease of vacancy mobility because the vacancy concentration increases as δ increases. At 296°C (and down to at least 250°C) the diffusivity drops again for large δ , particularly along the *c* direction. These conditions correspond to the monoclinic structure and Bertaut's vacancy superlattice. The vacancies segregate preferentially to alternate basal planes and tend to avoid being nearest neighbors. On the basis of these observations it is concluded that short-range ordering (clustering) of the vacancies reduces their mobilities and long-range ordering reduces the mobilities still more.

Bertaut (1953) also has observed structural modifications in FeS at low temperatures that lead to supercell reflections. Fasiska (1972) investigated FeS and Fe_{0.5}S₄ and concluded that at high temperatures the iron vacancies are nearly random but that close-packed layers become pleated by small displace-

ments of sulfur atoms and that twins and stacking faults are common consequences of quenching.

With respect to geological problems that depend on stoichiometric ratios, it is important that composition changes can be effected by the motion of vacant sites which are much more mobile than the ions; that is,

$$C_{\text{Fe}} D_{\text{Fe}} = C_{\text{S}} D_{\text{S}}$$

where C 's are concentrations and $C_{\text{Fe}} \gg C_{\text{S}}$. When D_{S} is 10^{-8} cm²/sec, distances of about 1 cm can be homogenized in times of about 15 years. Thus a compositional gradient would have to be present on a coarse scale or the pyrrhotite cooled rapidly on a geological scale to prevent homogenization above a few hundred degrees centigrade. Very large

pressures are required for moderate decreases in diffusivities.

SUMMARY

Mechanisms of diffusion in several sulfides have been reviewed in order to show that diffusive transport in crystals depends upon the crystal structure and especially upon their point defects, their concentrations, and their arrangements.

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